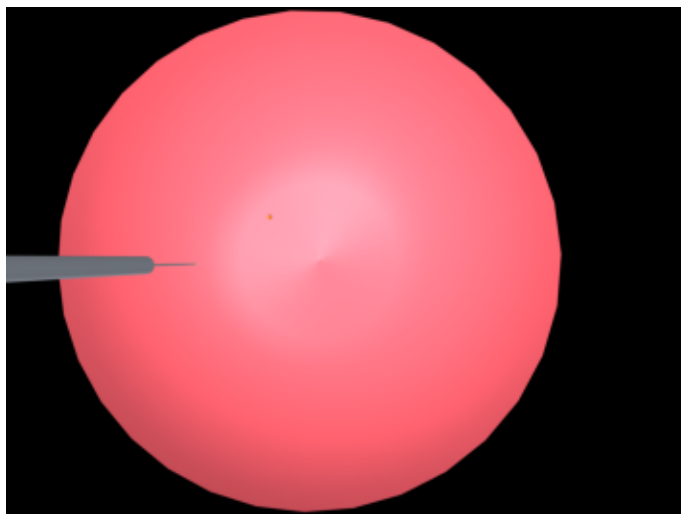
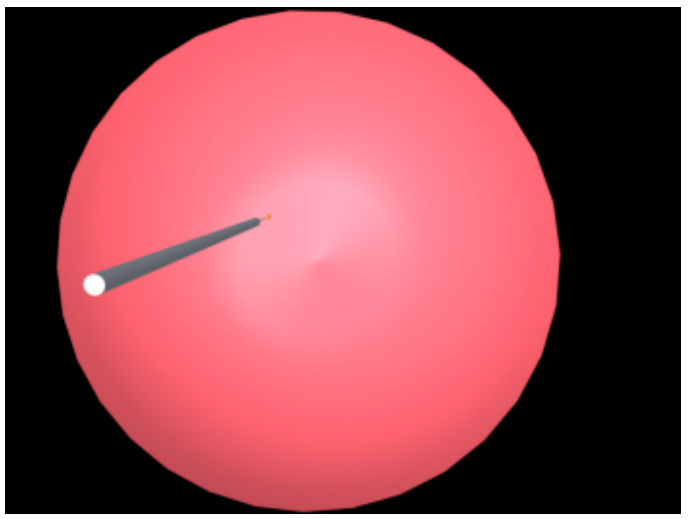
Raw controller RGB frames (uncropped; display scaling only; no overlays)

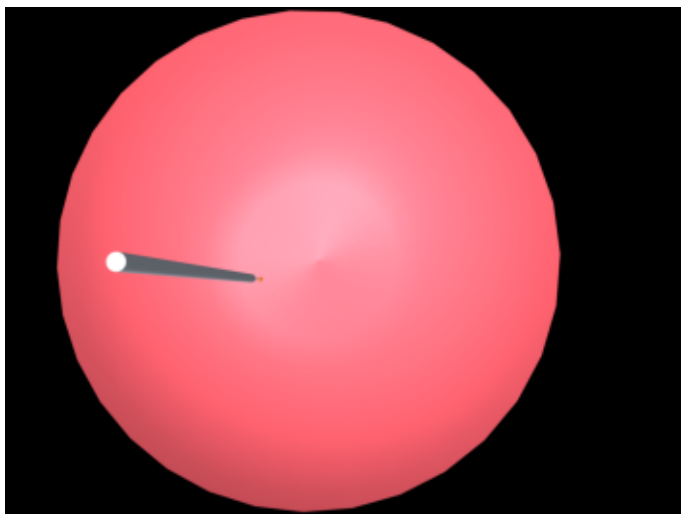
Hybrid control / P3 / t=0.00s
start / clear



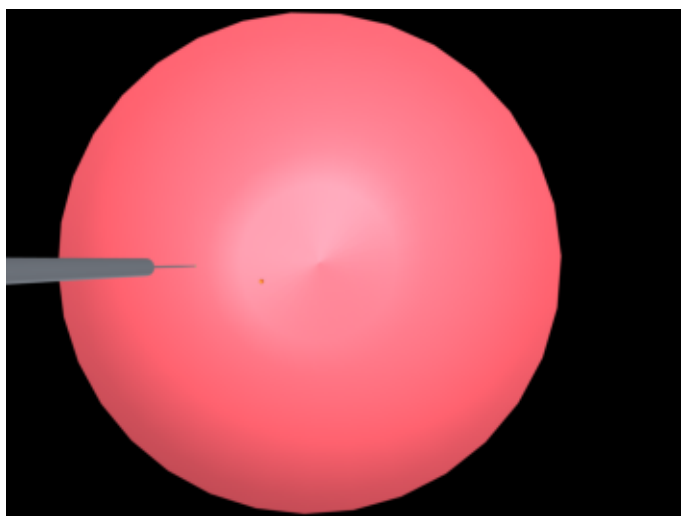
Hybrid control / P3 / t=31.69s
natural partial needle occlusion / partial_needle



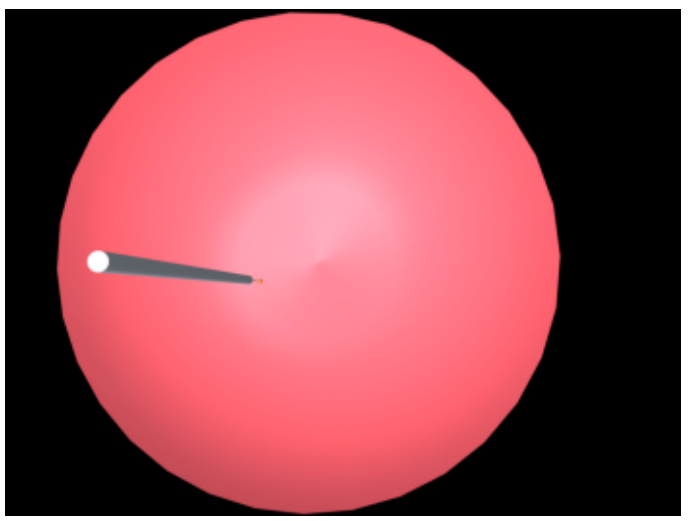
Hybrid control / P1 / t=57.12s
final recorded frame / partial_needle



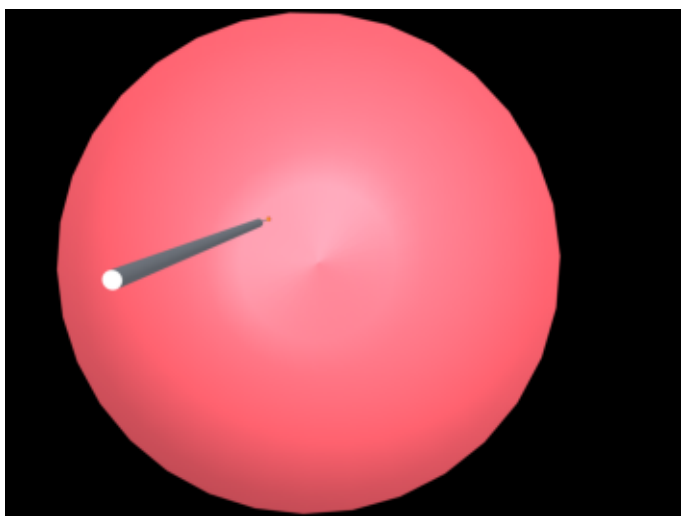
Strict RGB-D / P1 / t=0.00s
start / clear



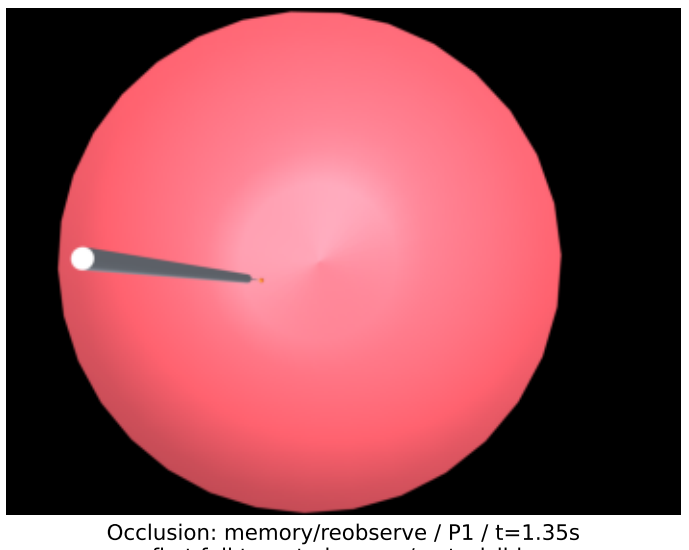
Strict RGB-D / P1 / t=34.45s
natural partial needle occlusion / partial_needle



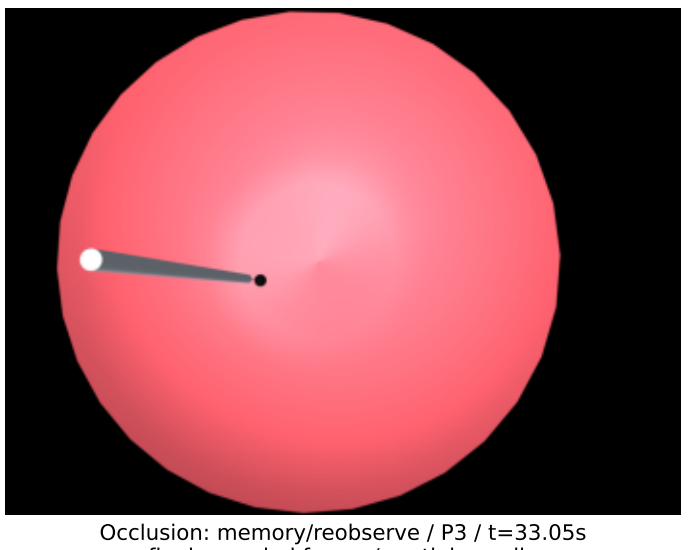
Strict RGB-D / P3 / t=64.05s
final recorded frame / partial_needle



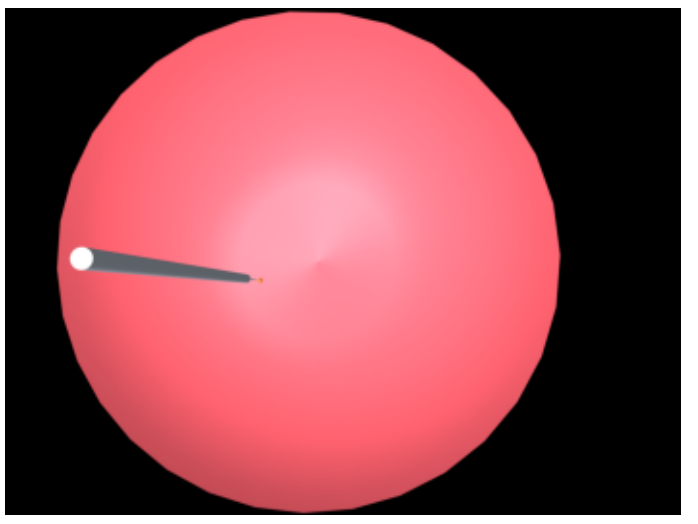
Occlusion: stop / P1 / t=0.00s
start / clear



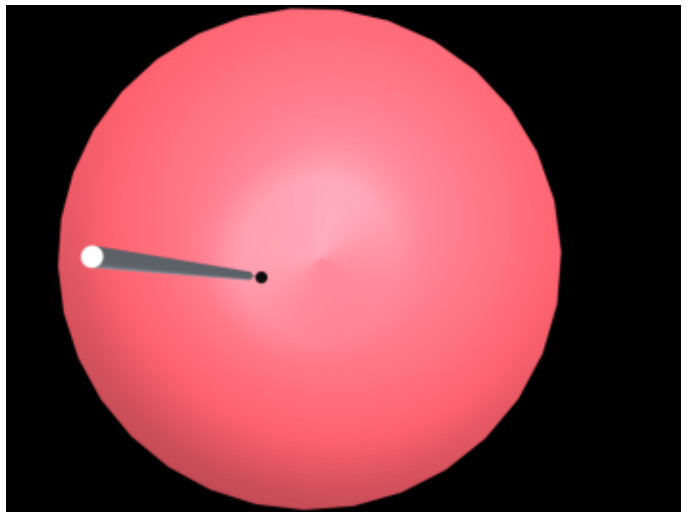
Occlusion: stop / P1 / t=1.35s
first full target absence / not_visible



Occlusion: memory/reobserve / P1 / t=0.00s
start / clear



Occlusion: memory/reobserve / P1 / t=1.35s
first full target absence / not_visible



Occlusion: memory/reobserve / P3 / t=33.05s
final recorded frame / partial_needle

